1(a) operations in 8086

A 1000

Give following commands without address

Address	Mnemonics
1100	MOV AX, 2
1103	MOV BX, 2
1106	ADD AX, BX
1108	MOV [1200], AX
1116	HLT

G 1000

Enter

Reset

M 1200

MULTIPLICATION

```
MOV DX,0002
MOV AX, 0002
MOV BX, 02
MUL BX
MOV DI, 1520
MOV [DI], AL
INC DI
MOV [DI], AH
INC DI
MOV [DI], DX
HLT
```

2)largest Number

A 1100

Address	Label	Mnemonics
1000		MOV SI,2000H
1003		MOV CL, [SI]
1005		MOV CH,00H
1007		INC SI
1008		MOV AL, [SI]
100A		DEC CL
100C		INC SI
100D	L2:	CMP AL,[SI]
100F		JNC/JC , L1 (1013)

```
1011            MOV AL,[SI]
1013    L1:     INC SI
1014            LOOP L2 (100D)
1016            MOV [2100],AL
101A            HLT
G1100
Sd 1013
```

Give input size (4)
Give 4 inputs by clicking space button for every number
After 4th number give "enter"
Sd 2100

3(BASIC ARITHMETIC OPERATIONS USING 8051)
Addition
A enter
Give 8500
enter

```
Address   Mnemonics
8500      MOV A, 2
8502      ADD A, 2
8504      MOV DPTR, #8600
8507      MOVX @DPTR, A
8508      HERE: SJMP HERE
Go 8500
enter
reset
Sd 8600
```

Multiplication

```
Address   Mnemonics
8500      MOV A, #06H
8502      MOV F0, #03H
8505      MUL AB
8506      MOV DPTR, #8600
8509      MOVX @DPTR, A
850A      INC DPTR
850B      MOV A,F0
```

850D MOVX, @DPTR, A
850E SJMP HERE //here=850E give current address

4(Stepper Motor)

Address	opcode	Mnemonics
1100	B080	MOV AL,80
1102	BA 36 FF	MOV DX,FF36
1105	EE	OUT DX,AL
1106	BE 00 12	START:MOV SI,1200
1109	B3 04	MOV BL,04
110B	8A 04	REPEAT:MOV AL,[SI]
110D	BA 30 FF	MOV DX,FF30
1110	EE	OUT DX,AL
1111	E8 07 00	CALL DELAY
1114	46	INC SI
1115	FE CB	DEC BL
1117	75 F2	JNE REPEAT
1119	EB EB	JMP START
111B	B9 03 09	DELAY:MOV CX,0903
111E	49	LOOP:DEC CX
111F	75 FD	JNE LOOP
1121	C3	RET

After this
1)Reset
2)M1200
3) enter four values
4) 03 [give space]
5) 06 [give space[
6) 0C[GIVE SPACE]
7) 09 [give double space]
8)reset
9) G1100[enter] motor will run

5)ADC
Give A[give enter]
Give 4100

```
MOV DPTR, #FFC8
MOV A, #10
MOVX @DPTR, A
MOV A, #18
MOVX @DPTR, A
MOV DPTR, #FFD0
MOV A, #01
MOVX @DPTR, A
MOV A, #00
MOVX @DPTR, A
MOV DPTR, #FFD8
MOVX A, @DPTR
JNB E0, WAIT[wait=4115]
MOV DPTR, #FFC0
MOVX A, @DPTR
MOV DPTR, #4150
MOVX @DPTR, A
SJMP HERE[here=4121]
Enter
Reset
Go 4100
Executing….
Reset
Adjust potentio meter
Press red button
Reset
Go 4100
Sd 4150
```

Format (led bulb) 1 2 4 8 8 4 2 1

6) TRAFFIC LIGHT CONTROLLER

CODE to write

```
void setup() {
pinMode(A0, OUTPUT);
pinMode(A1, OUTPUT);
pinMode(A2, OUTPUT);
Serial.begin(9600);
}
```

```
void loop() {
digitalWrite(A0, HIGH);
digitalWrite(A1, LOW);
digitalWrite(A2, LOW);
Serial.println("STOP");
delay(5000);
digitalWrite(A0, LOW);
digitalWrite(A1, HIGH);
digitalWrite(A2, LOW);
Serial.println("Ready");
delay(2000);
digitalWrite(A0, LOW);
digitalWrite(A1, LOW);
digitalWrite(A2, HIGH);
Serial.println("Go");
delay(5000);
}
```

7)Basic Program using Arduino

LED Blink

```
void setup() {
pinMode(LED_BUILTIN, OUTPUT);
}
void loop() {
digitalWrite(LED_BUILTIN, HIGH);
delay(1000);
digitalWrite(LED_BUILTIN, LOW
delay(1000
}
```

LED Fade

```
const int ledPin = 9;
const int fadeRate = 5;
void setup() {
pinMode(ledPin, OUTPUT);
}
void loop() {
for (int brightness = 0; brightness <= 255; brightness++) {
analogWrite(ledPin, brightness);
delay(fadeRate);
```

```
}
for (int brightness = 255; brightness >= 0; brightness--) {
analogWrite(ledPin, brightness);
delay(fadeRate);
}
}
```